DEVELOPMENT OF MACRO-LEVEL CRIME AND COLLISION PREDICTION MODELS TO SUPPORT DATA DRIVEN APPROACH TO CRIME AND TRAFFIC SAFETY (DDACTS)

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ABSTRACT

The number of violent crimes and fatal-injury collisions concerns many jurisdictions. Traditional enforcement tactics are often reactive, relying on historical crime and collision data to select locations for law enforcement. Advanced law enforcement tactics take a proactive approach. Such tactics include Data-Driven Approaches to Crime and Traffic Safety (DDACTS) which uses predicted numbers of crimes and collisions to identify locations for law enforcement. Our DDACTS study was conducted in Regina, Saskatchewan, Canada. We developed macro-level prediction models to predict violent crimes and collisions in each traffic analysis zone (TAZ) in Regina. The zonal nature of the analysis is important for overcoming confidentiality and privacy issues associated with violent crimes and fatal-injury collisions. We used 54 input variables to describe each TAZ's crimes, collisions, socio-demographic, road inventory, traffic, and land use characteristics. Our analysis used negative binomial regression coupled with the empirical Bayes method (a popular approach in transportation, but relatively new to crime mapping) to develop two statistical models that predict the long term mean value for the number of violent crimes/collisions per zone. We used cumulative residual plots as our main goodness-of-fit test. The findings are summarized on a map showing the top 10 hotzones for violent crimes, the top 10 hotzones for fatal-injury collisions, and the zones where the crime and collisions zones overlap. The overlapping zones are the DDACTS zones. By focusing law enforcement in the DDACTS zones, it may be possible to reduce violent crimes and fatal-injury collisions simultaneously and use limited resources more cost-effectively.

Keywords: Violent Crime, Fatal-Injury Collision, Macro-level Prediction, Law Enforcement, Data-Driven Approaches to Crime and Traffic Safety (DDACTS)

INTRODUCTION

Study Purpose

The purpose of this study is to develop sophisticated macro-level (zonal-level) crime and collision prediction models to support the advanced place-based law enforcement system known as "Data-Driven Approaches to Crime and Traffic Safety (DDACTS)." We used crime and collision data collected from Regina, Saskatchewan, Canada. Regina is the capital of Saskatchewan and had a population of 247,200 in 2016 ***(1)***. Various confidentiality and privacy issues are clearly associated with crime data. As it may not be possible to disclose, for example, the specific locations of some crimes, many crime studies have taken a macro-level approach ***(2,3,4)***.

Our study focuses on violent crimes and fatal-injury collisions, i.e., the most serious/severe types of crimes and collisions. By analyzing the locations of recorded crimes and collisions and using the analysis to predict the expected future locations of crimes and collisions (defined as hotzones in our study), the study is expected to enable Regina Police Service to concentrate their law enforcement resources in the way that is most likely to simultaneously and cost-effectively reduce the number of crimes and collisions.

Strategies for Law Enforcement

Traditional law enforcement strategies rely heavily on responding to calls and on random or sporadic patrols. These strategies are essentially reactive rather than proactive: police officers typically have to wait until an area is identified as having a crime or traffic collisions problem before taking law enforcement action. This reactive approach to law enforcement is likely to include intuitive and even random elements that may not be cost-effective ***(5)***.

In contrast, proactive enforcement systems aim to predict the occurrence of crimes and collisions and apply appropriate law enforcement measures to reduce the number and/or severity of crimes and collisions and even to prevent some crimes and collisions from taking place at all ***(6)***. Proactive enforcement systems include sophisticated intelligence-led policing, predictive policing, selective enforcement, and smart enforcement. Proactive approaches based on prevention through data supported intelligence are designed to increase cost-effectiveness especially if crimes and collisions can be reduced simultaneously ***(7, 8, 9)***.

Data-Driven Approaches to Crime and Traffic Safety (DDACTS)

DDACTS is a proactive enforcement system used in North America ***(10)***. Combined with highly visible traffic enforcement, DDACTS is regarded as a sustainable, flexible and cost-effective method for reducing crime and collisions ***(9)***. The system seeks to improve the effectiveness and efficiency of law enforcement operations by enabling police officers to conduct focused, highly visible, law enforcement policing tactics based on independent, scientific, and evidence-based crime and collision data analysis. Highly visible traffic enforcement in an area helps to reduce the public's fear of crimes/collisions in that area. It can also have a positive effect on adjacent neighborhoods by reducing crime and collisions in those neighborhoods (the spillover effect). As a DDACTS strategy aims to reallocate the areas targeted for law enforcement rather than change the overall frequency or duration of law enforcement, the budget effect is expected to be neutral: police officers will be assigned to certain locations for law enforcement with or without a DDACTS strategy ***(6)***.

The use of a DDACTS strategy provides a scientifically justifiable approach to the selection of areas for enhanced law enforcement. This is important for the local community, for local administrators and for the police force. Under the current approach, some members of the public may, for example, become suspicious of law enforcement and may complain of profiling or revenue-motivated ticketing ***(5)***. Without a clear justification for the areas targeted for law enforcement, it is difficult to counter these perceptions.

In the United States, DDACTS is recognized as an intelligence-led policing business model that can help police officers to increase the effectiveness of law enforcement in a cost-efficient way. The approach has become a valuable policing strategy. Jurisdictions currently successfully applying DDACTS include many metropolitan areas (e.g., Baltimore County, MD, and Nashville, TN) and many small towns (such as St. Albans, VT, and Sheboygan, WI) ***(11, 12)***. In Baltimore, for example, DDACTS was introduced in early 2009. The number of crimes in the three years before DDACTS was compared with the six-month period (April to November, 2009) after DDACTS was introduced. The reduction in crimes included a 1.1% reduction in thefts from motor vehicles, a 16.6% reduction in burglaries, a 33.5% reduction in robberies, a 40.9% reduction in motor vehicle thefts, and a 43% reduction in injury collisions ***(13)***.

Place-Based Law Enforcement

Past research has shown that there is a relationship between dangerous driving behavior (e.g., reckless or impaired driving) and criminal behavior. Researchers have developed a number of criminological and psychological studies and theories to explain the overlap between crimes and collisions ***(9,14,15)***. The studies suggest that individuals who are involved in risky behavior in one activity (e.g., offensive and antisocial activities) will likely be involved in risky behavior in other activities (e.g., excessive speeding or drug impaired driving). The studies also found that such individuals often exhibit more than one behavioral problem. For example, individuals with a lack of self-control may engage in persistent risk-taking behavior ***(9,16)***.

Some crimes/collisions do not have identifiable causation motives or factors, but most crimes/collisions have a known place. Crimes and collisions (problems) and law enforcement (fixing problems) therefore have an important geographic element ***(17)***. Many recent studies in geographic criminology show that some places attract individuals who commit crimes and/or traffic violations (some of the traffic violations presumably lead to collisions). Such places are typically "rough" neighborhoods with high-crime rates. They may be considered typical examples of a geographic criminology theory known as deviant place theory ***(18)***.

From these perspectives, the idea of hotzone areas where crime and collisions overlap becomes very likely and a statistical analysis using, for example, a DDACTS approach can be expected to show the hotzones. The idea that the hotzones can be predicted also seems likely. It is also reasonable to expect that law enforcement that concentrates on the crime/collision overlap areas will likely reduce crimes and collisions at that location simultaneously.

It is not uncommon for law enforcement agencies to use GIS techniques to create crime maps or collision maps and then inspect the maps for locations showing a high concentration of crimes or collisions. Unfortunately, however, simple visual inspection can easily be misleading as the determination of genuine hotzones is not straightforward. For example, if we use a history of crime/collision occurrence numbers, the hotzones are likely to move and may change from year to year complicating the identification of areas where crimes and collisions consistently overlap. This phenomenon occurs due to regression-to-the mean bias ***(19,20,21)***. Regression-to-

the-mean refers to a statistical tendency. In our case, it is important because locations may experience an extreme value for crimes/collisions in one year and then, without any intervention, experience less extreme values in subsequent years. To avoid the pitfalls of regression-to-the-mean bias in crime/collision data analysis, enforcement agencies need maps based on the long term mean value of crimes/collisions. The creation of these maps requires rigorous spatial data analysis that uses appropriate and advanced statistical modeling techniques.

During the last two decades, there have been intensive discussions regarding the development of better-fitting crime/collision prediction models using a wide range of advanced statistical methods ***(22)***. The negative binomial regression model coupled with the empirical Bayes (EB) technique is currently regarded as one of the most popularly using methods for developing collision prediction models ***(23,24,25,26,27)***, but the current standard approach to mapping DDACTS zones uses kernel density analysis based simply on the observed number of crimes and collisions. In this study, we use a negative binomial regression model coupled with the empirical Bayes (EB) technique to develop predict the number of crimes and collisions. We do this to mitigate against regression-to-the-mean bias and we believe we are the first DDACTS study to do so. We developed different models for the number of violent crimes and for the number of fatal-injury collisions.

STUDY OBJECTIVES

The Regina DDACTS study had two main objectives:

- Develop macro-level crime and collision prediction models using an advanced statistical technique; and
- Demonstrate how the Regina Police Service can use the models' results to identify the hotzones where significant numbers of violent crimes and fatal-injury collisions overlap and where law enforcement should be focused (i.e., DDACTS zones).

STUDY DATA

The study used five years (2009 to 2013) of Regina crime, collision, traffic volume, socio-demographic and land use data to develop the crime and collision prediction models. We used traffic analysis zones (TAZ) as the spatial analysis unit. There were 299 traffic TAZs in Regina. The data were supplied by three different agencies:

- Five years (2009 to 2013) of crime data (GIS shapefile format) were provided by Regina Police Service;
- Five years (2009 to 2013) of collision data (Microsoft Access format) were supplied by Saskatchewan Government Insurance (SGI);
- Five matching years (2009 to 2013) of traffic volume data (GIS shapefile format) were supplied by the City of Regina;
- Roadway inventory data (GIS shapefile format) were supplied by the City of Regina. The data included the location of every intersection and roadway segment in the city (i.e., the roadway network) and details such as the functional classification and speed limit of every roadway segment; and
- Socio-demographic data (Microsoft Excel format) for each TAZ in Regina were supplied by the City of Regina.

During the five-year study period, there were 18,232 violent crimes and 5,759 fatal-injury collisions. The violent crime data refer to five specific categories of crime (assault, arson, robbery, sexual assault, and murder).

TABLE 1 provides an overview of the study's 54 input variables (two response variables (i.e., violent crimes and fatal-injury collisions) and 52 explanatory variables). All 54 input variables are aggregated to give a single-value for each TAZ. Note that the City of Regina chose and categorized the socio-demographic variables when the City collected data for each TAZ. The City's main purpose was to forecast traffic volumes on their road network. Our study used the same data, but recognized that the City of Regina's data choices might not be ideal for our study purpose. For example, we might have preferred some additional roadwork variables and different age and education categories.

TABLE 1: Description of Input Variables

Input Variable	Description (per Traffic Analysis Zone)	Mean	Std. Dev	Min	Max
Crime Data					
Violent Crimes	Total Number of Violent Crimes	69.06	563.37	0.00	9,115.90
Collision Data					
Fatal-Injury Collisions	Total Number of Fatal-Injury Collisions	21.98	21.65	0.00	103.00
Socio-Demographic Data					
POP_01to17	Population of Persons Aged 1 to 17	161.00	179.45	0.00	782.00
POP_18to24	Population of Persons Aged 18 to 24	80.00	84.12	0.00	339.00
POP_25to44	Population of Persons Aged 25 to 44	215.00	220.58	0.00	884.00
POP_45to64	Population of Persons Aged 45 to 64	204.00	214.47	0.00	911.00
POP_65plus	Population of Persons Aged 65 or above	104.00	125.84	0.00	763.00
TOT_POP	Total Population	765.00	785.32	0.00	3,011.00
POP_DENSITY	Population Density (Persons/km^2)	2,006.34	1,665.67	0.00	10,552.61
NO_GRDSCH	Number of Persons Enrolled in Graduate School	112.00	261.35	0.00	1,720.00
Road Inventory Characteristics					
AVE_SPDLIM	Weighted Average Speed Limit (km/hr)	85.03	55.84	40.00	536.17
TOT_ROADLEN_20	Roadway length with posted speed 20km/hr	0.76	12.33	0.00	200.00
TOT_ROADLEN_30	Roadway length with posted speed 30km/hr	4.50	45.01	0.00	523.00
TOT_ROADLEN_40	Roadway length with posted speed 40km/hr	636.09	818.10	0.00	3,836.75
TOT_ROADLEN_50	Roadway length with posted speed 50km/hr	3,147.48	2,703.98	0.00	14,405.84
TOT_ROADLEN_60	Roadway length with posted speed 60km/hr	28.57	151.33	0.00	1,695.00
TOT_ROADLEN_70	Roadway length with posted speed 70km/hr	122.42	392.76	0.00	3,284.75
TOT_ROADLEN_80	Roadway length with posted speed 80km/hr	130.25	431.64	0.00	3,677.00
TOT_ROADLEN_100	Roadway length with posted speed 100km/hr	70.10	251.46	0.00	1,820.00
TOT_ROADLEN	Total Road length, m	4,140.16	3,299.99	87.00	16,354.95
AVE_ROADLEN	Weighted Average Roadway Length, m	2,150.34	1,828.17	81.17	11,361.34
NO_3LEGS_INT	Number of Intersections with 3 Legs	9.02	10.02	0.00	66.00

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TABLE 1: Description of Input Variables (Cont'd)

Input Variable	Description (per Traffic Analysis Zone)	Mean	Std. Dev	Min	Max
Road Inventory Characteristics (Cont'd)					
NO_4LEGS_INT	Number of All Intersections with 4 Legs	5.21	6.41	0.00	36.00
NO_5LEGS_INT	Number of All Intersections with 5 Legs	0.02	0.12	0.00	1.00
TOTAL_INT	Total Number of Intersections	14.22	13.63	0.00	79.00
INT_DENSITY	Intersection Density, (Number of intersections per TLKM)	3.19	1.79	0.00	18.35
ARTERIAL_LEN	Arterial Length, m	555.93	565.05	0.00	2,618.06
COLLECTOR_LEN	Collector Length, m	621.16	751.80	0.00	3,452.70
GRAVEL_ROAD_LEN	Gravel Road Length (m)	136.50	488.31	0.00	4,879.00
EXPRESSWAY_LEN	Expressway Length, m	84.11	262.64	0.00	2,029.00
HIGHWAY_LEN	Highway Length, m	41.26	205.32	0.00	1,820.18
LOCAL_ROAD_LEN	Local Road Length, m	2,369.14	2396.95	0.00	9,643.52
PRIVATE_ROAD_LEN	Private Road Length, m	235.89	995.59	0.00	12,726.00
RAMP_LEN	Ramp Length, m	90.96	275.05	0.00	1,825.64
ROW_LEN	Right of way Length, m	4.10	32.73	0.00	434.00
TLKM	The sum of Length of all road segments in a TAZ, km	4.14	3.30	0.09	16.35
I3WP	Number of 3-leg intersections as percentage of total number of intersections, %	54.38	34.22	0.00	100.00
ALKP	Arterial road length as percentage of total-lane-kilometers, %	19.82	23.85	0.00	100.00
LLKP	Local road length as percentage of total-lane-kilometers, %	47.38	30.64	0.00	100.00
Traffic Volume					
VKT	Vehicle-Kilometer-Traveled	23,899.17	21,490.66	43.43	122,800.88
Land Use					
COMMERCIAL_AREA	Total commercial area, m^2	28,150.48	46,800.05	0.00	312,623.00
INSTITUTIONAL_AREA	Total institutional area, m^2	16,050.20	46,778.39	0.00	620,487.35
OPENSPACE_RECREATION_AREA	Total area designated for open space/recreation purposes, m^2	93,818.46	248,948.94	0.00	2,501,580.25
RAILWAY_AREA	Total area designated for railway use, m^2	10,560.52	52,694.82	0.00	691,835.00
RESIDENTIAL_HD_AREA	Total high density residential area, m^2	16,016.33	32,603.35	0.00	196,674.39
RESIDENTIAL_LD_AREA	Total low density residential area, m^2	172,483.88	210,194.97	0.00	786,855.00

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TABLE 1: Description of Input Variables (Cont'd)

Input Variable	Description (per Traffic Analysis Zone)	Mean	Std. Dev	Min	Max
Land Use (Cont'd)					
RESIDENTIAL_MD_AREA	Total medium density residential area, m^2	13,932.13	43,650.73	0.00	383,555.00
URBAN_HOLDING_AREA	Area intended to protect lands required for future urban developments, m^2	83,510.43	322,005.57	0.00	3,150,675.91
OFFICE_AREA	Office Floor Space Area (m^2)	3,431.05	10,266.25	0.00	101,970.00
RETAIL_AREA	Retail Floor Space Area (m^2)	6,192.72	13,029.34	0.00	122,199.00
INDUSTRY_AREA	Industry Floor Space Area (m^2)	6,134.17	24,540.63	0.00	274,205.00
HOSPT_SPACE	Hospital Floor Space (m^2)	445.75	4,732.10	0.00	66,704.00
URBAN_HOLDING_RESIDENTIAL_AREA_PROP	Proportion of TAZ Area designated as Urban Holding Residential Area	0.17	0.34	0	1.00
NO_LU_PER_TAZ	Numbers of Different Land Uses per Traffic Analysis Zone	4.13	1.68	1.00	8.00
TAZ_AREA	Area of Traffic Analysis Zone, m^2	521,414.91	570,457.13	0.00	6,044,326.67

METHODOLOGY

Negative Binomial Regression Model

Our study used negative binomial (NB) regression in the development of the crimes and collision prediction models. The functional form of the NB model employed in this study is given as Equation (1) for the expected mean (μ_i) which is then re-expressed in Equations (2) and (3) for each zone i:

$$\ln(\mu_i) = x_i^T\beta + \varepsilon_i \quad (1)$$

$$\mu_i = exp\left(x_i^T\beta + \varepsilon_i\right) \quad (2)$$

$$y_i \sim x_i^T\beta \quad (3)$$

where, y_i is the observed value of the response variable (i.e., the number of violent crimes or fatal injury collisions) for zone i,

x_i is the explanatory variable for zone i, and;

β is the coefficient for zone i.

y_i follows a negative binomial distribution. The probability density function for the NB model can be written as Equation 4 with dispersion parameter α.

$$P\left(y_i|\mu_{i,}\alpha\right) = \frac{\Gamma\left(\frac{1}{\alpha} + y_i\right)}{\Gamma(y_i+1)\Gamma\left(\frac{1}{\alpha}\right)}\left(\frac{1}{1+\alpha\mu_i}\right)^{\frac{1}{\alpha}}\left(\frac{\alpha\mu_i}{1+\alpha\mu_i}\right)^{y_i} \quad (4)$$

The likelihood function of Equation (4) is:

$$L(\mu|y,\alpha) = \prod_{i=1}^{n} \exp\left\{y_i \ln\left(\frac{\alpha\mu_i}{1+\alpha\mu_i}\right) - \frac{1}{\alpha}\ln(1+\alpha\mu_i) + \ln\Gamma\left(\frac{1}{\alpha}+y_i\right) - \ln\Gamma(y_i+1) - \ln\Gamma\left(\frac{1}{\alpha}\right)\right\} \quad (5)$$

The log likelihood function of the NB model is shown in Equation (6):

$$\ln(\beta|y,\alpha) = \sum_{i=1}^{n} y_i ln\left(\frac{\alpha exp(x_i^T\beta)}{1+\alpha exp(x_i^T\beta)}\right) - \frac{1}{\alpha} ln\left(1+\alpha exp(x_i^T\beta)\right) + ln\Gamma\left(\frac{1}{\alpha}+y_i\right) - ln\Gamma(y_i+1) - ln\Gamma\left(\frac{1}{\alpha}\right) \quad (6)$$

We used R-Language ***(28)*** to estimate the parameters of the NB models developed.

Empirical Bayes Method

Regression-to-the-mean bias is a common issue in collision prediction modeling. To overcome this bias, we used the EB method ***(29,30,31)***.

The EB method in Equation (7) uses information (i.e., NB model outputs) from similar zones to estimate the long term mean values (i.e., the EB adjusted values) of the number of crimes/collisions in a certain zone ***(26,32)***:

$$E[y] = w \cdot \mu_i + (1-w)y_i \quad (7)$$

where, $E[y]$ represents the EB adjusted value of violent crimes or fatal-injury collisions,

μ_i represents the predicted value of violent crimes or fatal-injury collisions,

y_i represents the observed value of violent crimes or fatal-injury collisions, and

w is the EB weight factor which is given in Equation (8):

$$w = \frac{1}{1+\alpha \times \sum_{t=1}^{Y} \mu_{ti}} \quad (8)$$

In Equation (8):

α is the dispersion parameter,

μ_{ti} is the predicted number of violent crimes or fatal-injury collisions for zone *i* in year *t*, and

Y is the 5 years in our study.

RESULTS OF THE ANALYSIS

Negative Binomial Regression Model

We began the analysis by developing a correlation matrix for the 54 input variables (the matrix is not presented in this paper due to its large size). The correlation matrix helped to highlight the strong relationships between the response variables and the explanatory variables. Explanatory

variables with strong relationships provide potentially strong candidates for producing better fitting results. For instance, POPULATION_25TO44 had the strongest linear relationship (r = 0.48) with crimes, and Vehicle-Kilometer-Traveled had the strongest linear relationship (r = 0.88) with collisions. We expected these two variables to contribute to improving the fitting performance of the crime and collision prediction models respectively.

Using the correlation matrix, we selected the explanatory variables for the crime and collision prediction models following a forward variable selection procedure, i.e., we added one variable at a time and checked whether the additional variable helped to improve the fitting performance. We did not use the most common confidence level (95%) when selecting statistically significant variables for our crime model or our collision model. Hauer ***(33)*** has argued that although a variable may not be statistically significant at 95% confidence level, the variable may still improve the fit and, presumably, the model's predictive performance. As a result, we added a variable to our crime and collision models if the variable contributed to improving predictive performance.

Various goodness-of-fit (GOF) tests are available to compare the predictive performance of candidate models ***(34,35)***. The tests include Mean Squared Error, Mean Squared Prediction Error, Mean Prediction Bias, and Mean Absolute Deviation. Residuals are used as an input value for examining a model's fitting performance for all the GOF tests listed. Hauer ***(33)*** emphasized the importance of applying cumulative residual (CURE) plots as a main GOF test. CURE plots show the magnitude of the residuals and their standard errors along the range of important explanatory variables (e.g., POPULATION_25TO44 and Vehicle-Kilometers-Traveled in our study). In this study, we used CURE plots as our main tool for understanding the predictive performance of the crime and collision prediction models we developed.

TABLE 2 shows the parameter values of the explanatory variables used in the two models and the associated p-values (i.e., Pr>|Z|). The prediction model for violent crimes included six input variables: COMMERCIAL_AREA, RETAIL_SPACE, and four population related variables. Commercial and retail areas are associated with business activity, retail stores and shopping malls, and with the related business and shopping trips. Such locations are often targets for crimes such as robbery and assault, and show a positive association with violent crime. Increased population density also appeared to be associated with an increase in violent crime. The age of the population also appeared to play a role.

A zone with a relatively large population between the ages of 18 and 44 showed a more positive geographic association with violent crime than a zone with a relatively large population in the 45 to 64 age group. (Note, however, that this finding does not necessarily mean that it was the younger people who were committing the crimes in the zones affected.)

Four of the six input variables had a statistically significant relationship with violent crime at the 95% confidence level. The relationships between POPULATION_18TO24 and violent crime, and RETAIL_SPACE and violent crime were not statistically significant at the same confidence level.

TABLE 2: Models Developed for Violent Crime and Fatal-Injury Collision Prediction

Input Variable	Description (per Traffic Analysis Zone)	Estimate	Pr>|Z|
Violent Crimes			
Intercept		-0.189	0.195
POP_18to24	Population of Persons Aged 18 to 24	0.0048	0.1385
POP_25to44	Population of Persons Aged 25 to 44	0.0085	<0.001
POP_45to64	Population of Persons Aged 45 to 64	-0.0076	<0.001
POP_DENSITY	Population Density (Persons/km^2)	0.0002	0.0022
Ln(COMMERCIAL_AREA)	Total Commercial Area, m^2 (Log-Transformed)	0.574	<0.001
RETAIL_AREA	Retail Floor Space Area, m^2	0.112	0.0799
Dispersion (α)		0.9921	
Fatal-Injury Collisions			
Intercept		-3.73	< 0.001
TOT_ROADLEN_70	Roadway length with posted speed 70km/hr	0.0002	0.084
TOT_ROADLEN_80	Roadway length with posted speed 80km/hr	-0.0004	0.0157
NO_3LEGS_INT	Number of Intersections with 3 legs	-0.014	0.0317
NO_4LEGS_INT	Number of Intersections with 4 legs	0.0304	< 0.001
INT_DENSITY	Intersection Density, (Number of intersections per TLKM)	0.671	0.0017
Ln(COLLECTOR_LENGTH)	Collector Length, m (Log-Transformed)	-2.6	0.0039
Ln(VKT)	Vehicle-Kilometer-Traveled	0.515	< 0.001
URBAN_HOLDING_RESIDENTIAL_AREA_PROP	Proportion of TAZ Area designated as Urban Holding Residential Area	-2.72	< 0.001
Dispersion (α)		0.4641	

The prediction model for fatal-injury collisions prediction model included eight variables including variables related to speed and intersections. Seven of the eight variables were statistically significant at the 95% confidence level. The exception was road segments with a posted speed limit of 70 km/hr.

The results show some interesting relationships. Zones with longer VKT had a higher number of fatal-injury collisions. Road segments with a posted speed limit of 70 km/hr, four-leg intersections and intersection density are all positively associated with fatal-injury collisions. In Regina, roads with a 70 km/hr posted speed limit are usually minor arterials with many access points to shopping malls or retail stores, but no center medians. These features of the road may increase the number of fatal-injury collisions. Road segments with a posted speed limit of 80km/hr have a negative association with fatal-injury collisions. This result may appear unexpected, but in Regina, roads with an 80km/hr posted speed limit are usually major arterials with center medians. The center medians likely help to reduce the number of head-on collisions and possibly contribute to reducing fatal-injury collisions. 4-leg intersections can be assumed to have higher traffic volumes and more conflicts than 3-leg intersections leading to an increase in

fatal-injury collisions. Increased intersection density for a zone also contributes to an increase in fatal-injury collisions.

FIGURE 1 shows the CURE plots ***(36)*** for the violent crime and fatal-injury prediction models. POPULATION_25TO44 is used in the violent crimes CURE plot, and VKT is used in the fatal-injury collisions CURE plot. The purpose of the CURE plots is to understand the fitting performance of the models. A good CURE plot should move around the zero coordinate of the plot and should tend to converge at zero.

The results show that the two models are generally a good fit. In the case of the violent crime prediction model, however, the upward movement of the cumulative residuals on the right side shows that the model tends to underestimate the predicted values for violent crimes in zones with approximately 750 or more people aged between 25 and 44. The violent crimes models should therefore be used with caution for these particular zones. Omitted variable bias may account for the underestimation in the crime prediction model. We expect, however, that the underestimation will be reduced when the EB method is used. This is because the predicted value will then be combined with the (probably higher) number of observed violent crimes in the zones.

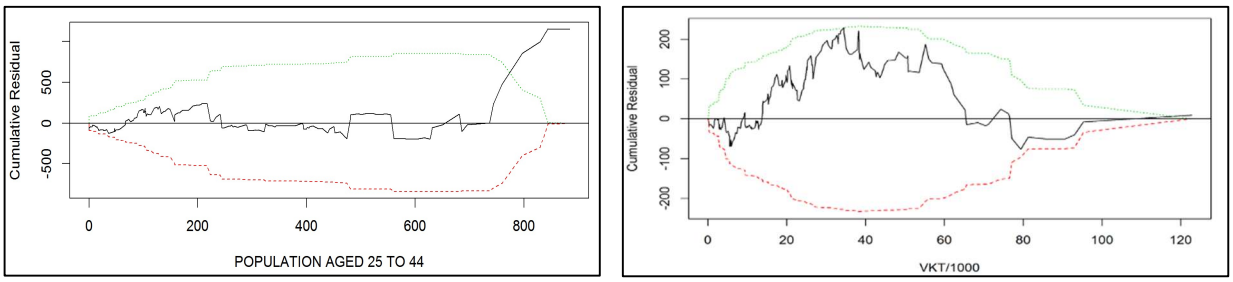


FIGURE 1: CURE Plots for Violent Crimes (Left) and Fatal-Injury Collisions (Right)

Crime Hotzones, Collision Hotzones and DDACTS Zones

The outputs listed in TABLE 2 were used as inputs to calculate the EB adjusted values of crimes and collisions (see Equation (7)). The calculation was repeated for each zone. Using the value produced, we were able to list the top 10 hotzones for violent crimes and the top 10 hotzones for fatal-injury collisions.

FIGURE shows the top 10 hotzones for violent crimes (blue) and fatal-injury collisions (red). The zones shown in purple are the zones where the violent crimes and fatal-injury collisions hotzones show a significant overlap. These are our four DDACTS zones.

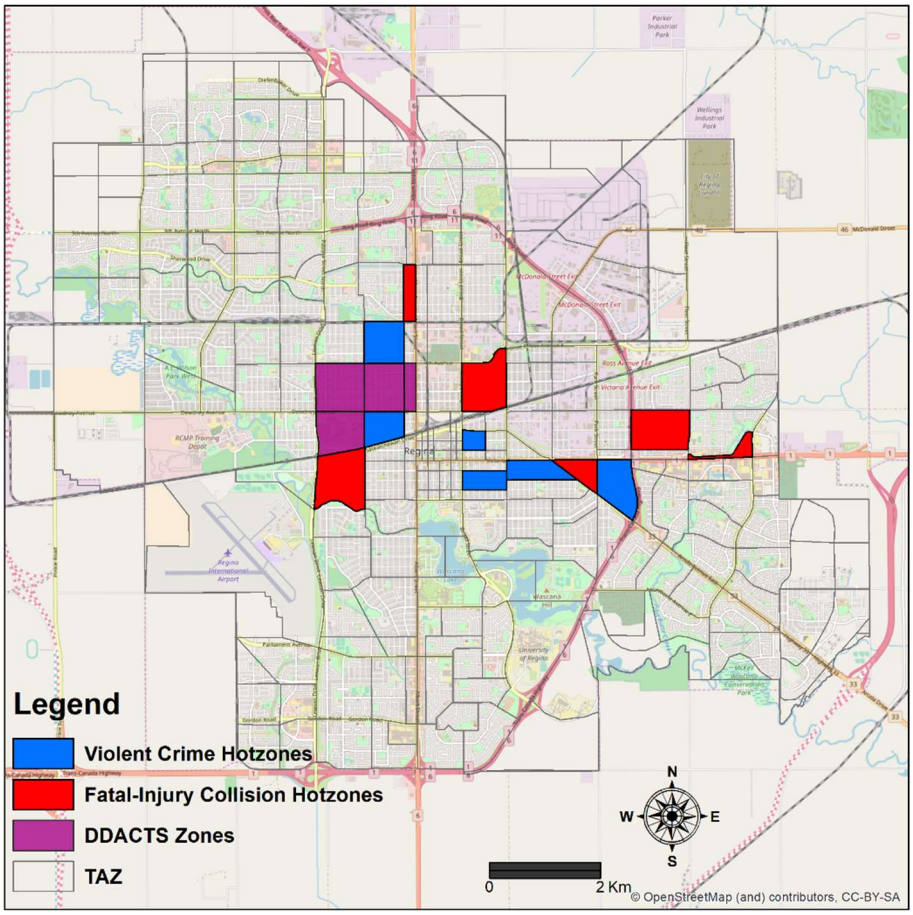


FIGURE 2: Crime Hotzones, Collision Hotzones and DDACTS Zones in Regina

TABLE 3 presents selected statistics for the top ten hotzones for violent crime and fatal-injury collision and for the four DDACTS zones. For instance, the top 10 violent crime hotzones account for 4.5% of Regina's area, but include 35.9% (3,278) of the city's violent crimes for the study period. The top 10 fatal-injury collision hotzones account for 5.2% of the city's area, but include 15.5% (892) of the fatal-injury collisions.

The DDACTS zones account for only 1.5% of Regina's area, but include 20.9% of the violent crimes and 6.1% of the fatal-injury collisions. These areas present a suitable focus for Regina Police Service's law enforcement and an opportunity to reduce violent crimes and fatal-injury collisions simultaneously.

330 **TABLE 3: Selected Statistics for the Top Ten Hotzones and DDACTS Zones**

Categories	Violent Crimes	Fatal-Injury Collisions
a. Top 10 zones' number of violent crimes or fatal-injury collisions during the study period	3,278	892
b. DDACTS zones' number of violent crimes or fatal-injury collisions during the study period	1,906	352
c. All zones' number of violent crimes or fatal-injury collisions during the study period	9,135	5,758
d. 'a' as a percentage of 'c'	35.9%	15.5%
e. 'b' as a percentage of 'c'	20.9%	6.1%
f. Area of top 10 zones	4.5 km^2	5.2 km^2
g. Area of DDACTS zones	2.2 km^2	2.2 km^2
h. Area of all zones	147.9 km^2	147.9 km^2
i. 'f' as a percentage of 'h'	3.0%	3.5%
j. 'g' as a percentage of 'h'	1.5%	1.5%

331 ## SUMMARY AND RECOMMENDATIONS FOR FUTURE WORK

332 This research started with two objectives: developing macro-level crime and collision prediction
333 models; and demonstrating how the models' results can be used to identify the hotzones where
334 significant numbers of violent crimes and fatal-injury collisions overlap. To meet these two
335 objectives, we collected five years (2009-2013) of macro-level data for Regina, Canada. We
336 considered 52 explanatory variables that described the socio-demographic, road inventory, traffic
337 volume, and land use characteristics for each zone of the study area. We also collected violent
338 crime and fatal-injury data for each zone. The number of violent crimes and the number of fatal-
339 injury collisions were our two response variables.

340 We conducted a correlation analysis to pre-screen potentially strong explanatory variables
341 for the violent crime and collision prediction models. We then used NB regression modeling (an
342 industry standard statistical method) to search for the best fitting models for crime and collision
343 prediction. We applied various GOF tests to assess the models, but as the residuals are used as an
344 input for all the GOF tests applied, we reported only the CURE plot results. The CURE plots
345 conveniently show the changes in magnitude of the residuals along the range of values of the
346 selected input variables.

347 The current standard approach to mapping DDACTS zones uses kernel density analysis
348 based on the observed number of crimes and collisions. Our study, however, used NB regression
349 modeling to predict the number of crimes and collisions and improve the prediction performance
350 of the NB models by using the EB method to produce the long-term mean value of the number of
351 crimes and collisions as the models' outcome. We believe our study is the first to try to reduce
352 regression-to-the-mean bias in the number of crimes and collisions concurrently and thereby
353 improve the selection of DDACTS zones for law enforcement.

The study results are presented on a map showing the top 10 hotzones for predicted violent crimes and fatal-injury collisions. The map also shows the four DDACTS zones where the violent crimes and fatal-injury collisions overlap and where it may be possible to focus highly visible traffic enforcement to reduce violent crimes and fatal-injury collisions simultaneously. This would enable the police service to increase the cost-effectiveness of their enforcement.

Future work may include the development of prediction models and hotzone maps that target other types of crime (e.g., auto theft or theft from auto) and other types of collisions (e.g., property damage only collisions). Additional information of this nature is likely to make further contributions to the cost-effectiveness of enforcement.

Future work may include other important factors. For example, weather conditions may have a significant influence on the occurrence of crimes and collisions. Analysis of complex factors such as weather conditions would require additional suitable explanatory variables some of which would need to be related to certain periods (e.g., seasonal traffic volume, number of shoppers at different seasons of the year, etc.). It may be relatively straightforward to relate weather conditions to existing crime and collision data, but developing and obtaining potentially important explanatory variables that vary with weather conditions is likely to be challenging.

Future work may also include a rigorous before and after study that evaluates how the adoption of DDACTS hotzones for increased police attention affects the number of crimes and collisions and their seriousness/severity.

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